INCH-POUND

MIL-R-5757/7E
6 March 1992
SUPERSEDING
MIL-R-5757/7D
18 February 1986

MILITARY SPECIFICATION SHEET

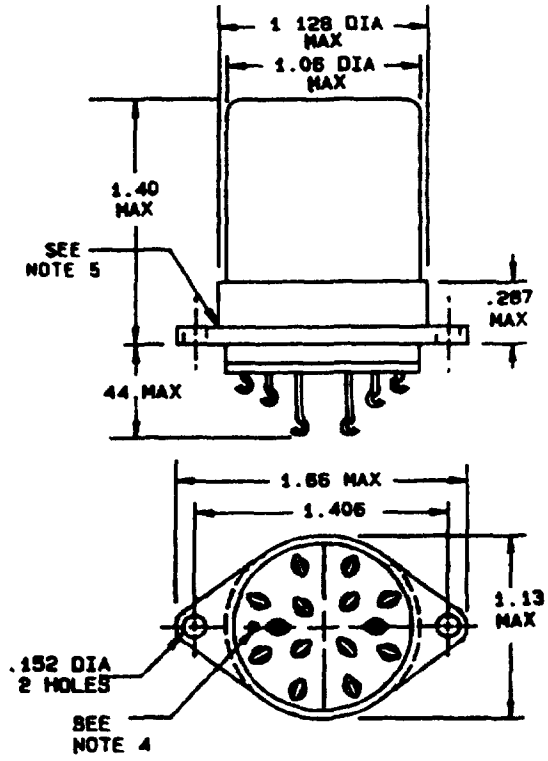
(E) RELAYS, ELECTROMAGNETIC, HERMETICALLY SEALED, 4PDT, LOW LEVEL TO 2 AMPERES

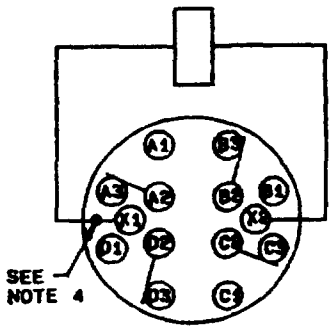
(E)

INACTIVE FOR NEW DESIGN AFTER 18 February 1986.
FOR NEW DESIGNS USE MIL-R-39016/14.

This specification is approved for use by all Departments and Agencies of the Department of Defense

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-R-5757.





CIRCUIT DIAGRAM
DEENERGIZED POSITION
TERMINAL VIEW

FIGURE 1. Relay - solder lug configuration (M5757/7-001)

(E) denotes changes

DISTRIBUTION STATEMENT A. Approved for public release, distribution is unlimited.

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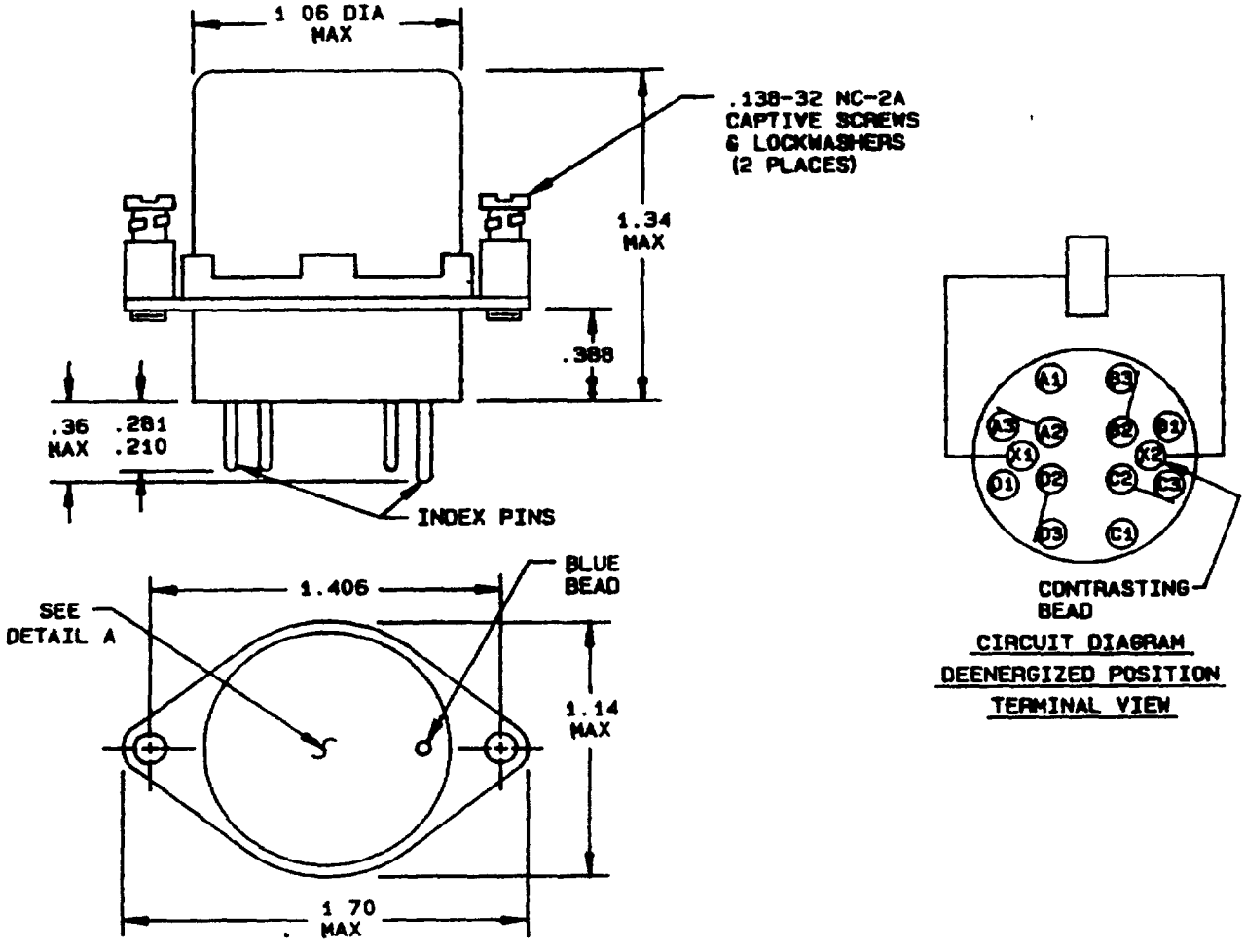
Inches	mm
.152	3.86
287	7.29
44	11.2
1.06	26.9
1.128	28.65
1.13	28.7
1.40	35.6
1.406	35.71
1 66	42 16

NOTES

1. Dimensions are in inches.
2. Unless otherwise specified, tolerance is ±.010 (0 25 mm).
3. Metric equivalents are given for general information only.
4. A colored line or dot shall be placed in the position shown, or optionally, the adjacent terminal in a clockwise direction around the outermost terminal circle, terminal number A3, shall have a contrasting bead.
5. Mounting screw head clearance shall be provided so that the relay may be mounted using a round head machine screw having .250 (6.35 mm) nominal head diameter. The mounting surface shall be flat with a minimum .276 (7.01 mm) diameter, and concentric with the mounting hole.

FIGURE 1. Relay - solder lug configuration (M5757/7-001) - Continued.

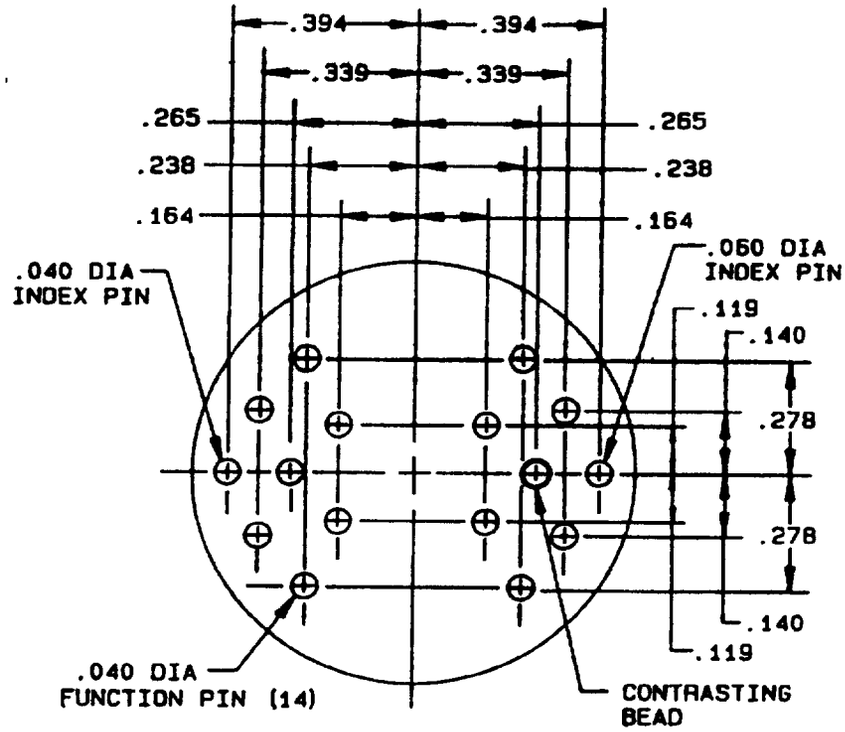
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Inches	mm	Inches	mm
.152	3.86	1.14	28.96
.287	7.29	1.40	35.6
.44	11.2	1.406	35.71
1.06	26.9	1.66	41.2
1 128	28.65	1.70	43.18

(E) FIGURE 2. Relay - socket pin configuration (M5757/7-003).

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DETAIL A

Inches	mm	Inches	mm	Inches	mm	Inches	mm
.040	1.02	.164	4 17	.281	7 14	1.06	26.9
.060	1.52	.203	5 16	.339	8 61	1 14	29.0
.085	2.16	.210	5 33	357	9.07	1.34	34.0
119	3.02	.238	6.05	36	9 1	1.406	35.71
138	3.51	.265	6.73	.388	9.86	1.69	42.9
140	3.56	.278	7.06	394	10.01		

NOTES:
1. Dimensions are in inches.
2. Unless otherwise specified, tolerance is ± 010 (0 25 mm)
3. Metric equivalents are given for general information only
4. All active electrical terminals shall be gold plated in accordance with MIL-G-45204, type II, class I, 0.00005 (50 microinches) inches minimum Gold plating of index pins is optional. Underplating, nickel 50 to 150 microinches.

FIGURE 2 Relay - socket pin configuration (M5757/7-003) - Continued

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REQUIREMENTS:

Contact data:

	High level characteristics (M5757/7-001)	Low level characteristics (M5757/7-003)
Configuration	4PDT	4PDT
Arrangement.	4 form C	4 form C
Load ratings (relay case grounded):		
Resistive:	2 amperes at 28 V dc	30 microamperes at 30 millivolts, 1000 Hz
Minimum current (resistive loads only)	2 amperes at 28 V dc 0.100 ampere at 28 V dc 0.500 ampere at 28 V dc	Not applicable
Contact resistance or voltage drop		
Rated life:		
Before	05 ohm maximum	.05 ohm maximum
During.	10 percent of open circuit voltage maximum	100 ohms maximum
After.	10 ohm maximum	15 ohm maximum
Minimum current		
Before:	.05 ohm maximum	Not applicable
During	3 ohms maximum	
After:	3 ohms maximum	
Contact bounce.	1 millisecond (ms) maximum	5 ms maximum
Overload.		
Resistive	Two times rated current	Not applicable

Coil data

Duty rating. Continuous

Maximum voltage: 32 V dc

Nominal voltage 26.5 V dc

Pickup voltage: 18 V dc maximum over temperature range.

Dropout voltage 14 V dc maximum over temperature range, 1.5 V dc minimum

Coil resistance.

-001. 225 ohms minimum, 280 ohms maximum at +25°C
-003 225 ohms minimum, 385 ohms maximum at +25°C

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Operate time 15 ms maximum

Release time 10 ms maximum

Electrical data

Insulation resistance 1,000 megohms minimum, except the resistance between coil and case at high temperature shall be 500 megohms or greater.

Dielectric withstanding voltage

	Sea level V rms (60 Hz)	Altitude V rms (60 Hz)
Between case, frame, or enclosure and all contacts both in the energized and deenergized position	1,000	350 - All terminals to case
Between case, frame, or enclosure and coil:	1,000	
Between all contacts and coil	1,000	
Between open contacts in the energized and deenergized positions·	500	
Between contact poles	1,000	

Environmental data

Temperature range Symbol B (-65°C to +125°C)

Vibration: Symbol 3 (15 g's, 10 to 2,000 Hz)

Acceleration: Applicable

Shock. Symbol 3 (100 g's).

Physical

Terminal strength. 5 ±0.5 pounds pull

Sealed by welding Optional

Dimensions and configuration See figures 1 and 2

Terminations. See figures 1 and 2.

Weight 0.2 pound maximum

Life test requirements

High level (applicable to M5757/7-001): 100,000 cycles at rated load

Low level (applicable to M5757/7-003): 100,000 cycles at rated load

Minimum current· Level II, 50,000 cycles

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Quality assurance:

Periodic inspection Groups B and C inspections are not applicable.

Group A.

a. Subgroup 1: Run-in (applicable to -003 low level relays only)

b. Subgroup 2: 100 percent.

c. Dielectric withstanding voltage.

(1) Tests to be conducted at sea level rating only.

(2) Duration of application: 5-10 seconds at a 10 percent increase in the dielectric strength voltage.

QUALIFICATION See table I or II as applicable.

TABLE I. Qualification inspection and sample size.

Part or Identifying Number (PIN) M5757/7	Number of sample units
-001	16 units plus 1 open unit. Qualification inspection as applicable
-003	12 units plus 1 open unit. Qualification inspection as applicable.

TABLE II. Extension of qualification and sample size

PIN M5757/7		Test	Number of sample units
Qualified	Extension to		
-001	-003	Low level life	4
-003	-001	High level life	4
		Minimum current	4
		Visual and mechanical	1

QUALIFICATION BY SIMILARITY: If the relay case, frame, or enclosure contains integral electromagnetic relays (mounting means excepted) currently listed on the qualified products list of MIL-R-39016, reduced testing shall consist of subjecting four sample units to the shock and vibration requirements of this specification sheet. Post tests shall include insulation resistance, dielectric withstanding voltage and electrical characteristics, and seal One unsealed unit shall be submitted to the qualifying activity.

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PIN Consists of the basic number of this specification sheet and a dash number

Example

M5757/7-001

Basic document number — Dash number

SUPERSESSION DATA: See table III

TABLE III Supersession data

PIN 1/ M5757/7	Supersedes	
	FSCM	PIN
-001	81349	RY4LB1A3L01
	81349	RY4LB1B3L01
	81349	RY4LB2A3L01
	81349	RY4LB2B3L01
	81349	RY4LB3A3L01
	81349	RY4LB3B3L01
-003	80063	SM-C-415281

1/ PIN M5757/7-002, type designations RY4LB1C3L01, RY4LB2C3L01, and RY4LB3C3L01 canceled without replacement

CONCLUDING MATERIAL

Custodians.
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AT, MI
Navy - SH
Air Force - 99
DLA - ES

User activities
Army - AV, ME

Preparing activity.
Navy - EC

Agent:
DLA - ES

(Project 5945-0874)

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